Laplace transform of unit step function

Express the following in terms of unit step function and find it's Laplace transform

1. $f(t)=\begin{cases}\cos t, 0<t<\pi\\ \cos 2t, \pi<t<2\pi\\ \cos 3t, t>2\pi\end{cases}$

2. $f(t)=\begin{cases}t^2, 0<t\le 2\\ 0, t>2\end{cases}$ ans: $\frac{2}{s^3}(1-e^{-2s})-\frac{4e^{-2s}}{s^2}(s+1)$

3. $f(t)=\begin{cases}t, 0<t\le 2\\ t^2, t>2\end{cases}$ ans: $\frac{1}{s^2}+\frac{e^{-2s}}{s^3}(2s^2+3s+2)$